Positive-Source Compression and All Schatten Classes for a Centered Theta Form

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Abstract

For the stated actual centered theta–Gamma–prime form, we construct a finite bank of centered images of compactly supported weighted polynomials that approximates the positive Fourier deficit of any finite prime head in trace. An explicit source rank gives the closed negative spectral count $N_-(\delta) \le \exp(O(\sqrt{1+\log(1/\delta)}))$, and the negative part of the centered operator belongs to $\mathfrak{S}_p$ for every $p > 0$, including trace class. Schubr previously obtained the same asymptotic count and all-Schatten conclusion for a realization whose even sector agrees with this one up to a factor of two. We give an alternative direct source-compression proof and an explicit, deliberately loose constant; this also sharpens the particular $\delta^{-1}\exp(O(\sqrt{\log(1/\delta)}))$ trace estimate of a later paper. The bank is used only in the ambient centered Hilbert space, not as a form- or operator-domain core. The result does not settle positivity, finite negative index, a gap at zero, or the Riemann hypothesis.

Keywords: theta kernel; prime-power form; source compression; negative spectral count; Schatten ideals.

1 Actual form and finite-head input

For $x \ge 0$ let $\tau_n(x) = \pi n^2 e^{2x}$ and set

$$\kappa(x) = e^{x/2}\sum_{n\ge1}(4\tau_n^2 - 6\tau_n)e^{-\tau_n}, \quad \kappa(-x) = \kappa(x), \qquad W(x) = \frac{\kappa(x)}{2\cosh(x/2)}.$$

Use the complex even Hilbert space $\mathcal{H} = L^2_{\mathrm{even}}(\mathbb{R}, dx/W)$ with conjugate-linear first argument, $m(u) = \int \cosh(x/2)u(x)dx$, $\mathcal{H}_c = \ker m$, and $Cu = u - 2m(u)\kappa$. The actual theta normalization makes C the orthogonal projection onto $\mathcal{H}_c$ and gives

$$\|\kappa\|_{\mathcal{H}} = 1, \quad B_0 := \int_{\mathbb{R}} W(x)dx \le \tfrac{1}{4}, \quad 0 < W(x) \le 40e^{4|x|-\pi e^{2|x|}} \le 40e^{4|x|-3e^{2|x|}}. \tag{1}$$

Let the ordinary Fourier transform be unitary. Define

$$\rho(s) = \frac{e^{-s/2}}{1-e^{-2s}}, \qquad d_\Gamma(t) = 2\int_0^\infty \rho(s)(1-\cos ts)ds \ge \tfrac{1}{2}\log(1+4t^2),$$

$$X_c = \left\{u \in \mathcal{H}_c : \int_{\mathbb{R}} d_\Gamma(t)|\widehat{u}(t)|^2dt < \infty\right\}, \quad q_0[u] = \int_{\mathbb{R}} d_\Gamma(t)|\widehat{u}(t)|^2dt - c_\Gamma\|u\|_2^2 - p[u],$$

where $c_\Gamma = \log\pi - \psi(1/4) \in (0,6)$ and

$$p[u] = 2\,\mathrm{Re}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\int_{\mathbb{R}} u(x+\log n)\overline{u(x)}\,dx.$$

The series includes every prime power and both orientations through its real part; its convergence and form representation are part of the cited input. We use as inputs the exact theta normalization, the closed lower-bounded form q_0 on X_c, and its form-defined self-adjoint representative L_c on $\mathcal{H}_c$, together with the complete prime-tail estimate, as established in the accompanying working manuscript and preceding trace paper [1, 2]. No equality with a separately defined minimal symmetric closure is asserted. The Gamma lower bound follows directly from $1-e^{-2s}\le 2s$, and the special value of ψ is standard [7].

For integer $N \ge 1$, write

$$c_N(t) = c_\Gamma + 2\sum_{2\le n\le N}\frac{\Lambda(n)}{\sqrt{n}}\cos(t\log n), \quad C_N = c_\Gamma + 2\sum_{2\le n\le N}\frac{\Lambda(n)}{\sqrt{n}},$$
$$a_N = (d_\Gamma - c_N)_+, \quad b_N = (c_N - d_\Gamma)_+, \quad T_N = \tfrac{1}{2}\sqrt{e^{2C_N}-1} > 0.$$

The scalar b_N is nonnegative, even, vanishes for $|t|\ge T_N$, and satisfies

$$\int_0^{T_N} b_N(t)dt \le T_N - \tfrac{1}{2}\arctan(2T_N) \le T_N. \tag{2}$$

For $g_t(x) = W(x)\cos(tx)/\sqrt{2\pi}$, the actual positive trace-class deficit on $\mathcal{H}_c$ is

$$B_N = 2\int_0^{T_N} b_N(t)(Cg_t)(Cg_t)^* dt, \qquad q_0[u] = \int a_N(t)|\widehat{u}(t)|^2 dt - \langle u, B_N u\rangle - \langle u, D_{>N}u\rangle \tag{3}$$

for every $u \in X_c$. The complete centered tail is bounded self-adjoint and

$$\|D_{>N}\| \le \tau_N \le 160(N+1)^2 e^{-\pi(N+1)}. \tag{4}$$

The full-domain form identity follows from finite-head Plancherel and the entire prime-tail norm convergence; the last two terms retain their actual signs. In particular,

$$q_0[u] \ge -\langle u, B_N u\rangle - \tau_N\|u\|_{\mathcal{H}}^2 \qquad (u \in X_c). \tag{5}$$

For $\delta > 0$ define the *closed* negative count

$$N_-(\delta) = \dim \mathrm{Ran}\,\mathbf{1}_{(-\infty,-\delta]}(L_c),$$

counting eigenvalues at $-\delta$ and multiplicities. A publicly posted preceding trace paper [2], which uses the same actual input realization and classical trace-count principle, proves $N_-(\delta) \le \delta^{-1}e^{O(\sqrt{\log(1/\delta)})}$ and $(L_c)_- \in \mathfrak{S}_p$ for $p > 1$. Earlier, Schubr [3, Theorem 1.1] obtained a subpolynomial negative-count estimate and all positive Schatten exponents for a constrained weighted Weil operator. Section 2 explains the even-sector form identification. The argument below is an alternative proof with an explicit constant, not a claim of first discovery or of a better constant.

2 Relation to earlier work

Schubr's publicly posted edition 3 [3], Theorem 1.1, gives a subpolynomial negative spectral count and membership of the negative part in every $\mathfrak{S}_p$, $p > 0$, for the complete constrained operator H defined there. We record the connection with the present form, rather than claiming those consequences as first results here.

In [3, Section 2, equations (4)–(6)], the source ψ is our κ, the weight $w = \psi \cosh(x/2)$, $s^2 = \psi/\cosh(x/2) = 2W$, the multiplier a equals d_Γ, and the scalar m_0 there is $-c_\Gamma$. The unitary map $U : \mathcal{H} \to L^2_{\mathrm{even}}(dx)$ is $Uu = u/\sqrt{W}$. For even u, the second constraint of [3] is automatic by parity, while its first is $\langle \sqrt{w}, Uu\rangle = \sqrt{2}\, m(u)$. Thus $U\mathcal{H}_c$ is its even constrained space. On centered even compact-smooth functions, the full form of [3] at Uu is exactly $2q_0[u]$: $sUu = \sqrt{2}u$, both complete prime orientations are retained, and its nonunitary Fourier transform with $dt/(2\pi)$ matches our unitary convention.

The form closures agree under the stated source inputs. Multiplication by $\sqrt{W}$ or its reciprocal maps compact smooth even functions bijectively to compact smooth even functions because W is smooth and strictly positive. The ambient compact-smooth form core from [1] yields a centered compact-smooth core: after approximating $u \in X_c$, subtract from each approximant a scalar multiple of one fixed even compact-smooth test with nonzero m; the scalar tends to zero by continuity of m. This is precisely the transformed even constrained core in [3]. Consequently the even restriction of its form-defined operator is unitarily equivalent to $2L_c$. The full constrained operator there reduces under parity, so its negative-count and all-Schatten estimates imply the corresponding conclusions for L_c; the use of strict thresholds there versus the closed threshold here does not alter the asymptotic, since the closed L_c threshold $-\delta$ is contained in its strict range below $-\delta/2$, corresponding to the strict H threshold $-\delta$. The explicit numerical constant 40 below is one admissible bound, not a claim to improve the constants of [3].

3 A centered source bank and its true Gram

Fix $R > 0$ and integer $m \geq 0$. In $\mathcal{H}_c$ set

$$v_j = \frac{1}{\sqrt{2\pi}} C\Big(W(x)x^{2j}\mathbf{1}_{[-R,R]}(x)\Big), \quad 0 \leq j \leq m, \quad V = (v_0, \dots, v_m), \quad r = m + 1. \tag{6}$$

Each vector belongs to $\mathcal{H}_c$, and the bank is independent. Indeed, if $C(Wp\mathbf{1}_{[-R,R]}) = 0$ for an even polynomial p, the uncentered vector is a scalar multiple of κ. Outside the interval it vanishes while $\kappa > 0$, so the scalar is zero; then $W > 0$ on the interval forces $p = 0$. This proof applies to complex coefficients.

Let

$$a_j = \int_{-R}^{R} \kappa(x)x^{2j}dx, \qquad w_\ell = \int_{-R}^{R} W(x)x^{2\ell}dx, \qquad G_{ij} = \langle v_i, v_j\rangle = \frac{w_{i+j} - a_i a_j}{2\pi}. \tag{7}$$

Thus $G = V^*V$ is positive definite as a matrix. The integrals in (7) are over the indicated *finite* interval, and the centering terms are retained. Define

$$P = VG^{-1}V^*, \qquad Q = I - P \quad \text{on } \mathcal{H}_c.$$

This is the true weighted orthogonal projection, not a coefficient-identity Gram or a projection onto the physical kernel. The source bank is required only to lie in $\mathcal{H}_c$. We do not require or use $v_j \in X_c$, $v_j \in \mathrm{Dom}\, L_c$, or preservation of either domain by P or Q.

4 Trace defect and an explicit rank

Theorem 4.1. *For every $\varepsilon > 0$, with $T = T_N$, define*

$$h = \max\{1, T/\varepsilon\}, \quad R = \tfrac{1}{2}\log\log(80h), \quad r = \left\lceil \max\left\{1, 3TR, \tfrac{1}{2}\log_2(2h)\right\}\right\rceil, \quad m = r - 1. \tag{8}$$

For the bank (6) *and Q above,*

$$\operatorname{tr}(QB_NQ) < \varepsilon, \qquad r \le 3TR + \tfrac{1}{2}\log_2(2h) + 2. \tag{9}$$

Proof. Set $k = 2m + 2 = 2r$ and $p_{m,t}(x) = \sum_{j=0}^{m}(-1)^j(tx)^{2j}/(2j)!$. For $|x| \le R$ and $0 \le t \le T$, Taylor's theorem to order $2m + 1$ gives

$$|\cos(tx) - p_{m,t}(x)| \le E := \frac{(TR)^k}{k!}.$$

The centered truncated Taylor vector belongs to $\operatorname{Ran} P$. Since C is contractive, the projection distance, with disjoint inside and outside regions, satisfies

$$\|QCg_t\|_{\mathcal{H}}^2 \le \frac{W_{\mathrm{tail}}(R) + B_0E^2}{2\pi}, \qquad W_{\mathrm{tail}}(R) = \int_{|x|>R} W(x)dx. \tag{10}$$

Consequently, by (3) and (2),

$$\operatorname{tr}(QB_NQ) = 2\int_0^T b_N(t)\|QCg_t\|_{\mathcal{H}}^2 dt \le \frac{T}{\pi}\left(W_{\mathrm{tail}}(R) + B_0E^2\right). \tag{11}$$

This bound includes the full centered tail of each source: no uncentered truncation is treated as already centered.

For $Y = e^{2R} \ge 1$, the envelope in (1) has logarithmic derivative at most -2 on $x \ge 0$. Integrating the tangent exponential majorant on both half-lines yields

$$W_{\mathrm{tail}}(R) \le 40Y^2e^{-3Y} \le 40e^{-Y}. \tag{12}$$

With (8), $Y = \log(80h)$, so $W_{\mathrm{tail}}(R) \le 1/(2h)$. The factorial estimate $k! \ge (k/e)^k$, $e < 3$, and $k = 2r \ge 6TR$ give $E \le 2^{-k} \le 1/(2h)$. Using $B_0 \le 1/4$ in (11),

$$\operatorname{tr}(QB_NQ) \le \frac{T}{\pi}\left(\frac{1}{2h} + \frac{1}{16h^2}\right) \le \frac{9T}{16\pi h} \le \frac{9\varepsilon}{16\pi} < \varepsilon.$$

The rank bound follows from $\lceil \max(a, b, c)\rceil \le a+b+c+2$ for the displayed nonnegative quantities. □

For a certified finite implementation, use rigorous upper bounds for T, h, R and choose an even k satisfying the displayed inequalities strictly; if the radius is enlarged, recheck $k \ge 6TR$. This avoids deciding whether a computable real is exactly an integer. It does not provide a bound on the condition number of G, required arithmetic precision, or total runtime.

5 Negative count and Schatten classes

Theorem 5.1. *For $0 < \delta \le 1$, the actual centered negative spectral count satisfies*

$$N_-(\delta) < \exp\left(40\sqrt{1 + \log(1/\delta)}\right). \tag{13}$$

In particular $N_-(\delta) \le \exp(O(\sqrt{1 + \log(1/\delta)}))$ as $\delta \downarrow 0$. The constant 40 is deliberately loose.

Proof. Put

$$M = \max\{8, \lceil \log_2(2/\delta) \rceil\}, \qquad N = M - 1. \tag{14}$$

Then $\tau_N \le \delta/2$. Indeed, for $M \ge 8$,

$$160M^2 e^{-\pi M} \le e^{-\pi M/2} \le 2^{-M} \le \delta/2.$$

The first inequality holds at $M = 8$ because $160 \cdot 8^2 < (5/2)^{12} < e^{3 \cdot 8/2} < e^{\pi 8/2}$, and its ratio improves for $M \ge 8$; the second follows from $\pi/2 > \log 2$.

Use Theorem 4.1 for B_N with $\varepsilon = \delta/4$. If the closed negative spectral subspace had dimension greater than r, it would contain a nonzero $u \in \operatorname{Ran} Q$. Lower boundedness of L_c places that spectral subspace in $\operatorname{Dom} L_c \subset X_c$. Since $B_N \ge 0$,

$$\langle u, B_N u \rangle = \langle u, QB_NQu \rangle \le \operatorname{tr}(QB_NQ)\|u\|_{\mathcal{H}}^2 < \frac{\delta}{4}\|u\|_{\mathcal{H}}^2.$$

Then (5) gives $q_0[u] > -3\delta\|u\|_{\mathcal{H}}^2/4$, contradicting $q_0[u] \le -\delta\|u\|_{\mathcal{H}}^2$. Thus $N_-(\delta) \le r$. This argument uses only that P is a bounded finite-rank projection on $\mathcal{H}_c$; it never applies L_c to a bank vector or assumes $PX_c \subset X_c$.

For the displayed constant, the elementary full von Mangoldt estimate in [2] gives $C_N \le 16 + 6\sqrt{N}$. Put $L = 1 + \log(1/\delta) \ge 1$ and $x = \sqrt{L}$. From (14) and $\log 2 > 1/2$, $N < 8L$; hence $C_N < 34x$ and $T < e^{34x}$. With $h = \max\{1, 4T/\delta\}$, $\log h < 36L$, so

$$R = \tfrac{1}{2}\log\log(80h) < 2 + \tfrac{1}{2}\log L \le \tfrac{5}{2}x < e^{2x}.$$

Also $\frac{1}{2}\log_2(2h) < 37L$. The explicit rank formula gives

$$r < 3e^{36x} + 37L + 2 \le 42e^{36x} < e^{40x},$$

using $L = x^2 \le e^{36x}$ and $e^4 > 42$. This proves (13). The constants depend only on the displayed actual input estimates, not on a Gram condition number or an inverse computation. □

Corollary 5.2. *The negative part $(L_c)_- = (-L_c)_+$ is compact and belongs to $\mathfrak{S}_p$ for every $p > 0$, in particular to $\mathfrak{S}_1$.*

Proof. The count is finite at every positive threshold, and L_c is lower bounded. Its negative part is therefore the norm limit of finite-rank spectral truncations. If μ_n are its nonzero eigenvalues, the layer-cake formula for $p > 0$ gives

$$\sum_n \mu_n^p = p \int_0^{\|(L_c)_-\|} \delta^{p-1} \#\{n : \mu_n > \delta\}\, d\delta.$$

The strict count under the integral is bounded by the closed count (13). For any fixed $p > 0$, $e^{40\sqrt{1+\log(1/\delta)}} \le C_p \delta^{-p/2}$ near zero, and $\delta^{p/2-1}$ is integrable. For $0 < p < 1$, $\mathfrak{S}_p$ denotes the usual Schatten quasi-class. Trace class does not imply finitely many or no negative eigenvalues. □

6 Verification boundary and source positioning

The exact theta form realization, normalization, Gamma lower bound, and complete tail bound are input from [1, 2]; the supplied analytical notes [4, 5, 6] record the source-compression construction and its independent scope-limited review. The publicly posted preceding paper [2] established the weaker total-trace estimate and $p > 1$. Schubr [3] already established the all-positive-exponent consequence

for the equivalent even realization, as detailed in Section 2. The present finite-source defect argument is an alternative proof and improves the *specific* bound in [2]; it makes no first-result or better-constant claim. Standard positive trace and spectral-count methods are classical [8].

The bank projection is not a known-radical, full-kernel, or low-collar projection. No actual Gram has been numerically certified here. The source-rank bound does not control inverse-trial count, Gram conditioning, arithmetic precision, or runtime. No existence or absence of negative eigenvalues, finite negative index, zero gap, physical-kernel completeness, or Riemann hypothesis conclusion follows. The finite checks in the package address elementary algebra and constants, not the full form-realization inputs or a numerical sign.

References

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